

Animation Overdubbing: Animation Creating Tool for Non-humanoid Characters by Mixing Multiple Pose Remappings

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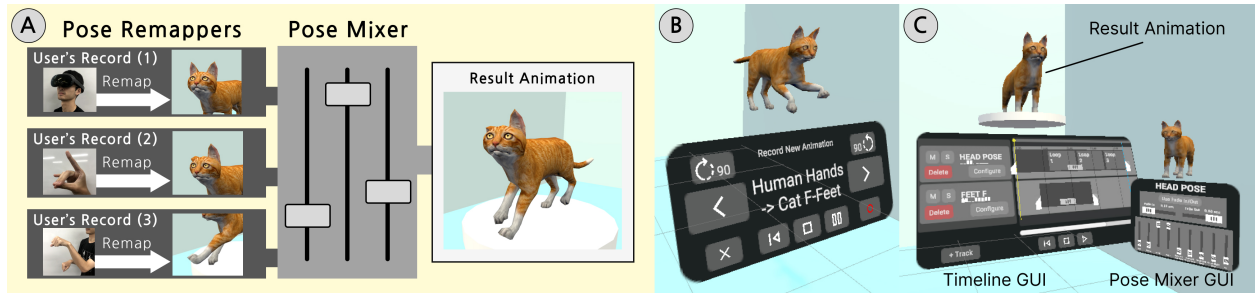


Figure 1: (A) The system of *Animation Overdubbing*. *Pose Mixer* mixes animations of non-humanoid characters recorded by *Pose Remappers*. (B) The GUI of the recording scene by *Pose Remappers*. The users select one of the *Pose Remappers* to control the non-humanoid character. (C) The timeline GUI. Each animation track on the timeline has a mixer GUI. These GUIs adjust the timing and the mixing weights of each animation.

Abstract

Creating animations for 3D non-humanoid characters presents several challenges because motion capture technology cannot be applied as effectively as it is for humanoid characters. We developed a new animation-creating system named “Animation Overdubbing”. This system creates an animation for a non-humanoid character by taking weighted averages of animations recorded by the users. The users can select one of the remapping methods when recording animations. By selecting multiple remapping methods, the users can control the characters with many degrees of freedom.

CCS Concepts

- **Human-centered computing** → **Virtual reality**; *Gestural input*;
- **Information systems** → *Multimedia content creation*.

Keywords

Virtual Reality, Non-humanoid Character, Motion Remapping, Motion Mixing

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1 Introduction

Creating characters' animation has a significant impact on economics and culture. It is known that creating 3D animation on 2D interfaces is laborious because 2D interfaces cannot fully utilize the degree of freedom in 3D space. Capturing actors' motion is a common approach to addressing this challenge. Recent studies have explored the creation of animations in a virtual environment [Bordier et al. 2022; Pan and Mitchell 2020] or an augmented environment [Ye et al. 2020]. In these studies, the user wears a head-mounted display and makes key-frames of characters in an immersed virtual environment.

Some characters that animators create have topologically different structures from humanoid characters. They are called non-humanoid characters. Creating animations for non-humanoid characters is harder and more time-consuming than for humanoid characters. This difficulty arises primarily because humanoid characters can be intuitively animated by human actors using their own body movements and motion capture devices, while non-humanoid characters cannot. To create animations of non-humanoid characters, animation creators still mostly rely on key-framing and interpolation. There are examples of high-quality animal animations created by attaching motion capture devices to animals, such as *Stray* [Interactive 2022]. However, this method is limited to real animals to which motion capture devices can be easily attached.

Several studies [Jiang et al. 2023; Seol et al. 2013] have addressed this challenge by remapping the poses of humanoid characters into those of non-humanoid characters in a virtual environment. These studies showed that human actors can potentially control non-humanoid characters using their bodies. However, these approaches have certain limitations. First, the degree of freedom (DoF) of the characters before and after remapping must be similar. When the

users use their DoF to control other functions such as recording the animation simultaneously [Fender et al. 2015], the limitation to the character would be more strict. Second, it is uncommon for human body parts to perfectly correspond to the functions of non-humanoid characters' body parts. For instance, cats use their mouths to eat and grasp objects, whereas humans do not. This discrepancy can lead to a lack of intuitive control. These issues arise because only one type of remapping is typically used. To address this, we propose a system called "Animation Overdubbing," which creates animations by blending multiple types of remapping applied to the same non-humanoid character.

2 System Overview

Our system consists of multiple Pose Remappers and a Pose Mixer as the main components, along with touchable GUIs in the virtual environment. A Pose Remapper converts the user's pose into the corresponding pose of a non-humanoid character (Figure 1 (A) and (B)). The remapping methods include inverse kinematics, joint matching, and other methods. In the inverse kinematics method, the non-humanoid character's limbs move based on inverse kinematics targeting the users' heads or hands. The joint matching method aligns the user's joints with those of the non-humanoid character, where the latter rotates correspondingly after an appropriate coordinate transformation. Our system includes multiple Pose Remappers in advance to control the whole body part of a non-humanoid character.

It is challenging to implement a Pose Remapper in a way that ensures the motion remains intuitive for the user, regardless of which body parts are matched. Therefore, we manually created Pose Remappers using several techniques. These techniques allow us to develop a variety of Pose Remappers while maintaining intuitive control over the non-humanoid characters. One technique in the inverse kinematics method is to place a transparent character at the user's position. The character is scaled in three directions so that the user's hands can reach the end-effector without having to move from his/her position. The whole pose of the transparent character is copied to a character in front of the user (Figure 1(B)), allowing the user to visually confirm the pose. Another technique used in the joint matching method is to define a constraint that preserves the hierarchical relationship of the joints before and after matching. This approach is based by HandAvatar [Jiang et al. 2023], which found that users felt more intuitive control over non-humanoid characters when this constraint was applied. For example, when moving a cat character's neck and ears, matching the user's right wrist and fingers is more effective than matching the user's right wrist and left fingers.

Users can instantly adjust Pose Remappers to ensure that the remapping methods align with their intuition. To select a Pose Remapper, users interact with the selector GUI (Figure 1(B)). Once a Pose Remapper is selected, users can start and stop recording the animations of non-humanoid characters that they control using their bodies and Pose Remappers, all through the same GUI. The recorded animations are then combined into a single non-humanoid character animation by the Pose Mixer (Figure 1 (A)). The Pose Mixer mixes several poses of a non-humanoid character into a single pose. When mixing, the Pose Mixer calculates the weighted average

of the joint angles. Users can configure the weights assigned to different groups of joints within the non-humanoid character. For example, users can blend two cat poses, one walking and the other shaking its head, into a single pose where the cat is both walking and shaking its head by adjusting the weights accordingly.

The users can configure the timing when each animation starts and finishes, and how many each animation repeats on the timeline GUI (Figure 1 (C)). They can also configure the mixing weights of the Pose Mixer with Mixer GUI. These GUI are designed based on existing software related to time series data such as video and music. The users can access the GUI by touching buttons and sliders with their tracked hands. The Users can assign all motions to character control since all system functions are accessed through the GUI.

3 Demo at SIGGRAPH Asia 2024

In the demonstration at SIGGRAPH Asia 2024, the participants create non-humanoid characters' animations via our system. We prepared several types of non-humanoid characters (see Figure 2).



Figure 2: Examples of non-humanoid characters used in our system. From left to right: cat, bird, dragonfly, ancient fish.

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References

- Jean-Baptiste Bordier, Anthony Mirabile, Robin Courant, and Marc Christie. 2022. Smart Motion Trails for Animating in VR. In *AIVR 2022 - 5th IEEE International Conference on Artificial Intelligence and Virtual Reality*. IEEE, Virtual, France. <https://inria.hal.science/hal-03935067>
- Andreas Fender, Jörg Müller, and David Lindlbauer. 2015. Creature Teacher: A Performance-Based Animation System for Creating Cyclic Movements. In *Proceedings of the 3rd ACM Symposium on Spatial User Interaction* (Los Angeles, California, USA) (SUI '15). Association for Computing Machinery, New York, NY, USA, 113–122. <https://doi.org/10.1145/2788940.2788944>
- Annapurna Interactive. 2022. Stray. <https://stray.game/>
- Yu Jiang, Zhipeng Li, Mufei He, David Lindlbauer, and Yukang Yan. 2023. HandAvatar: Embodying Non-Humanoid Virtual Avatars through Hands. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 309, 17 pages. <https://doi.org/10.1145/3544548.3581027>
- Ye Pan and Kenny Mitchell. 2020. PoseMMR: A Collaborative Mixed Reality Authoring Tool for Character Animation. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. 758–759. <https://doi.org/10.1109/VRW50115.2020.00230>
- Yeongho Seol, Carol O'Sullivan, and Jehée Lee. 2013. Creature Features: Online Motion Puppetry for Non-Human Characters. In *Proceedings of the 12th ACM SIGGRAPH/Eurographics Symposium on Computer Animation* (Anaheim, California) (SCA '13). Association for Computing Machinery, New York, NY, USA, 213–221. <https://doi.org/10.1145/2485895.2485903>
- Hui Ye, Kin Chung Kwan, Wanchao Su, and Hongbo Fu. 2020. ARAnimator: in-situ character animation in mobile AR with user-defined motion gestures. *ACM Trans. Graph.* 39, 4, Article 83 (aug 2020), 12 pages. <https://doi.org/10.1145/3386569.3392404>